

Richard B. Townsley

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U.Va. mechanical engineering graduate with hands-on experience in CAD design, ANSI-compliant fabrication, and operations coordination. Charlottesville-based and ready to apply skills in the field.

EDUCATION

University of Virginia

Bachelor of Science, Mechanical Engineering

Charlottesville, VA

May 2026

- **Relevant Coursework:** Mechatronics, Fluid Mechanics, Computational Methods (MAE), Computer-Aided Design (CAD), Strength of Materials, Heat Transfer, Machine Elements & Fatigue Design.

PROJECTS & RESEARCH

Radar-Actuated Mower Blade Guard | *Senior Capstone Design*

Aug. 2025 – May 2026

- Engineered a retrofit safety mechanism for riding mowers to prevent traumatic amputation injuries, adhering to ANSI safety standards.
- Designed mechanical assemblies in SOLIDWORKS and led rapid prototyping of components on a team of eight.
- Created a solenoid-actuated guard activated by millimeter-wave radar sensors and microcontrollers, eliminating the need for user input.
- Conducted testing and simulations to ensure reliable human detection without user input.
- Fabricated and tested a functional, cost-effective prototype (<20% of mower cost).

Reusable To-Go Program Ambassador | *U.Va Dine*

Jan. 2025 – May 2026

- Evaluated container materials to gauge integrity and resistance to impact/heat degradation.
- Developed life-cycle cost models in Julia to forecast waste reduction and financial impact.
- Surveyed 50+ students about check-out logistics and structural integrity.
- Translated findings into program recommendations to increase longevity and user adoption.

TECHNICAL SKILLS

Design & Modeling: SolidWorks (CSWA Certified - ID: C-ZMVFGP4HWE), CAD/CAM, GD&T, Technical Drawings

Fabrication: Machining (Lathe, Mill, Drill Press, Bandsaw), 3D Printing, Rapid Prototyping

Analysis & Programming: JavaScript, Julia (Data Analysis), MATLAB, Excel (Project Tracking)

Project Management: Project Scheduling (MS Project, Primavera), Cross-functional Collaboration, Technical Writing, Field Documentation

EXPERIENCE

University of Virginia (Aramark, U.Va. Dine)

Sep. 2024 – May 2026

Green Dining Intern (Operations & Logistics)

Charlottesville, VA

- Directed sustainability operations for 3 dining locations, managing 5,000+ reusable assets to minimize waste and optimize logistics.
- Coordinated with facilities management and external vendors to integrate new food options and manage events.
- Did weekly quantitative waste audits (100+ lbs of samples) to identify supply chain inefficiencies and change procurement decisions.

The Cavalier Daily

Feb. 2025 – May 2026

Staff Writer, Arts & Entertainment

Charlottesville, VA

- Independently pitched and authored analytical articles under strict deadlines, demonstrating strong written communication and project execution skills.
- Researched and interviewed local artists to produce content that engages a diverse community.

ADDITIONAL WORK HISTORY

Starbucks | *Barista, Clean Play*

Mar. 2024 – Aug. 2024

Sheetz, Inc. | *Store Team Member*

Mar. 2022 – Nov. 2022